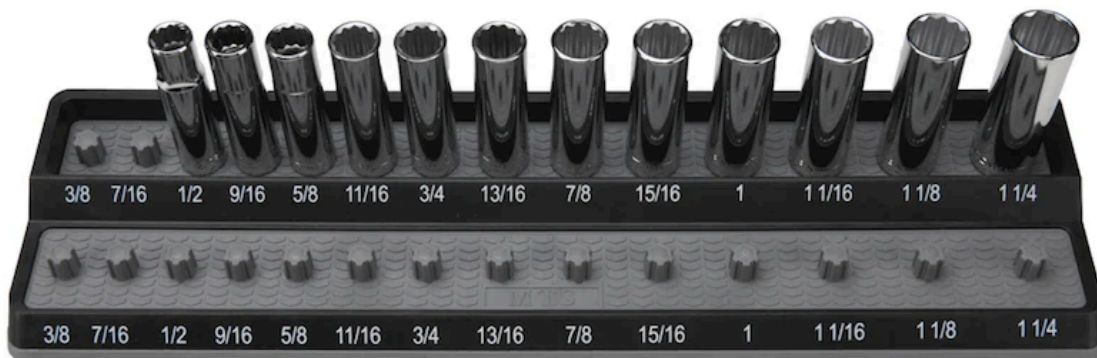


MyDesign Portfolio

Element B Initial Template

Element B: Documentation and Analysis of Prior Design Attempts

Prior Design Attempt #1 - <1/2" SAE Friction Peg Socket Holder>



-[mechanic time savers](#)

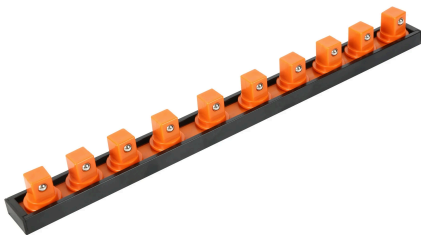
Stand alone, little slots for each socket size, two rows, label socket size.

It's inconvenient because it takes up a lot of space and won't be able to move to the workplace. It does not have a lot of mobility and grabbing it would be hard. The sockets could also fall off the pegs when being moved around or if dropped. Our stakeholder also uses square sockets so the star shapes will not fit our sockets.

Prior Design Attempt #2 -

<https://www.walmart.com/ip/Hyper-Tough-1-2-inch-Drive-Socket-Holder-for-Standard-and-Deep-Socket-Organizer-Model-5817/644901034?wl13=5227&selectedSellerId=0&wmlspartner=wlpa&StoreCode=5227&gQT=1>

Hyper Tough 1/2-inch Drive Socket Holder for Standard and Deep Socket Organizer



Details

A line of molds that can hold 10 sockets and can also be attached to a wall and still hold them. Is made out of plastic and the socket holders can also rotate around. Has an aluminum siding and weighs 4.2 oz.

Analysis

This design can only hold 10 sockets of a specific square size which does not fit what our stakeholder wants as he wants the ability to hold varying square sizes of sockets. It is in a line as it is exactly how our stakeholder wants. It also has a sturdy outside and is lightweight so it works for moving around without problems.



Prior Design Attempt #3 - <Plastic Based Horizontal Socket Holder>

https://www.google.com/imgres?q=socket%20organizer%20tray%20homemade&imgurl=https%3A%2F%2Fi.etsystatic.com%2F47552351%2Fr%2Fii%2Fe02846%2F6031580034%2Fii_1080xN.6031580034_fbsx.jpg&imgrefurl=https%3A%2F%2Fwww.etsy.com%2Flisting%2F1726758350%2Fstackable-universal-hex-or-torxs-mid&docid=jPD9765BFOGPM&tbnid=7_ay1bpVlw3BeM&vet=12ahUKewi2y9zozfOLAxVnFlkFHTp7BbsQM3oECCwQAA..i&w=1080&h=1440&hcb=2&ved=2ahUKewi2y9zozfOLAxVnFlkFHTp7BbsQM3oECCwQAA



Details

This attempt to solve the design problem is a Horizontal holder for sockets which holds 9 total including the shaft to use them.

Analysis



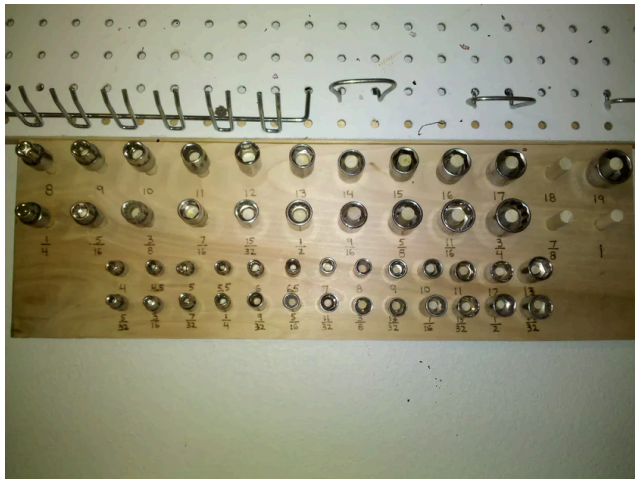
This design attempt is a good way to store sockets because it's clean and minimal, but the size of the design compared to the minimal number of sockets it can hold makes it unsuitable for the current problem at hand as we have around 42 sockets to hold in one drawer. This design, if fully incorporated, would only be able to hold around an estimated 36 sockets in the drawer with no excess space for other objects.

Prior Design Attempt #4 - <insert name of prior design attempt #4>

Source

. "A Better Socket Organizer." *Instructables*, Instructables, 27 Oct. 2017, www.instructables.com/A-Better-Socket-Organizer/.

Image



Details

This design is good because there is a pegboard with mounted socket wrenches and some hooks. The sockets are in two rows and we can see that the sockets will stay organized. To make that design you need Saw, Drill press, Wood glue and Wood Burning Kit (optional).

Analysis

The board on the wall is similar to what the stakeholder is currently using and he wants something that can be moved around easily. It also is not great as the way that he orders

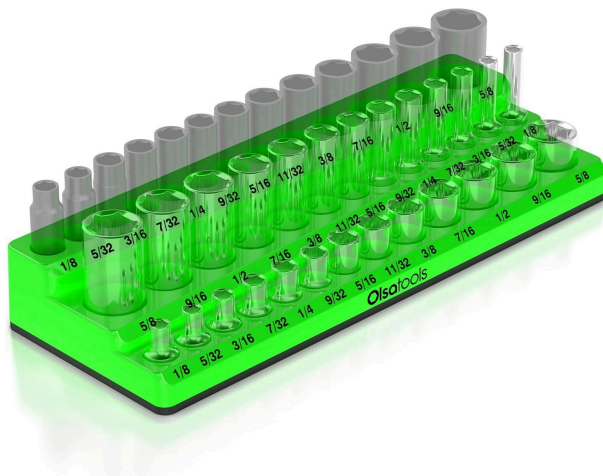
the sizes of the board is just based on the numbers below the pegs and makes it so that when putting sockets back it makes it so you could end up placing them in the wrong place.

Prior Design Attempt #5 - <insert name of prior design attempt #5>

Olsa Tools Magnetic Socket Organizer (1/4-Inch Drive, 3 Row) - Magnetic Tool Organizer & Socket Holder - Tool Box Socket Organizers - SAE Green - Holds 39 Sockets - Professional Grade

<https://www.amazon.com/Olsa-Tools-4-Inch-Magnetic-Organizer/dp/B09ZYX4T3N?source=ps-sl-s-hoppingads-lpcontext&ref=fplfs&smid=A1ZCMAA7R2I7TN&gQT=1&th=1>

Made by Olsa tools.



Details

This design is great as it features many parts that make it great to make sure your sockets stay organized. It has a magnetic bottom so that it can stay attached to magnetic surfaces to prevent it from moving around. Each hole is set to a specific dimension so that it won't have the sockets moving around loosely and falling out. It can hold a wide range of sockets also.

Analysis

While this prior design attempt did a lot of things right it is not quite right for us as our stakeholder wants it so that it is not large and unwieldy and does not need to hold a lot of sockets. He also wants it in strips and not one big part to hold all of the sockets.

Provide evidence and explain what parts of this prior design attempt were unsuccessful at solving the defined problem and why based on design requirements.